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Practice : 2

In [3]:
```
inputs = [1, 0, 2]
Wx, Wh, b = 0.6, 0.4, 0
h = 0
hidden_states = []

for x in inputs:
    h = (Wx * x) + (Wh * h) + b
    hidden_states.append(round(h, 3))

print(f"Hidden states = {hidden_states}")
```

```
Hidden states = [0.6, 0.24, 1.296]
```

In [4]:
```
ft, it, Ct_candidate, Ct_prev = 0.8, 0.5, 0.6, 1
Ct = (ft * Ct_prev) + (it * Ct_candidate)
print(f"New cell state = {round(Ct, 1)}")
```

```
New cell state = 1.1
```

In [5]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

model = Sequential([
    LSTM(5, input_shape=(3, 1)),
    Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam', loss='binary_crossentropy')

print("Model created successfully with:")
print("  LSTM layer (5 units)")
print("  Dense output layer (1 unit, sigmoid)")
model.summary()
```

```
Model created successfully with:
  LSTM layer (5 units)
  Dense output layer (1 unit, sigmoid)
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 5)	140
dense_1 (Dense)	(None, 1)	6

Total params: 146 (584.00 B)

Trainable params: 146 (584.00 B)

Non-trainable params: 0 (0.00 B)